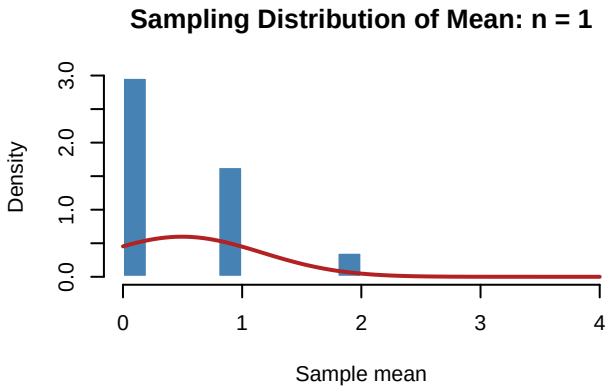
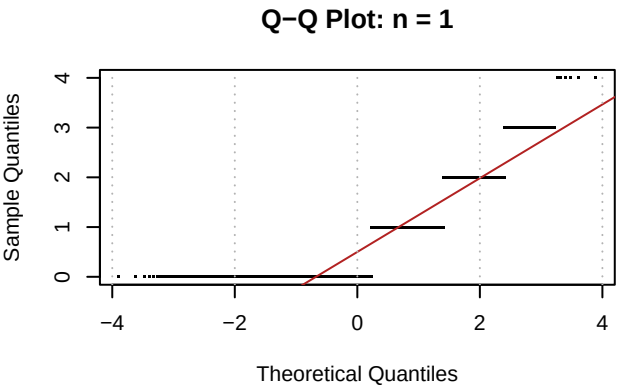
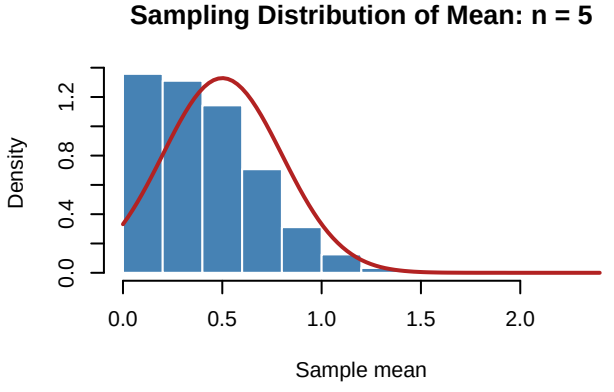
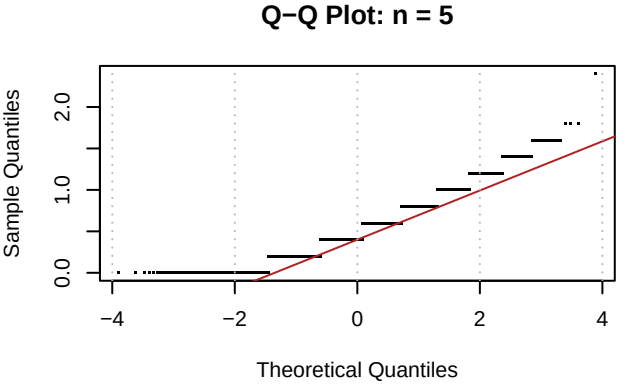
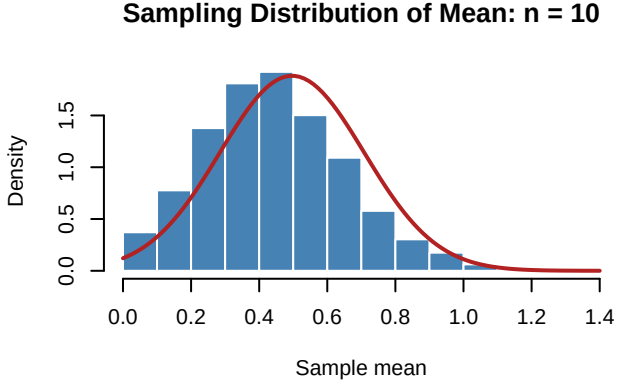
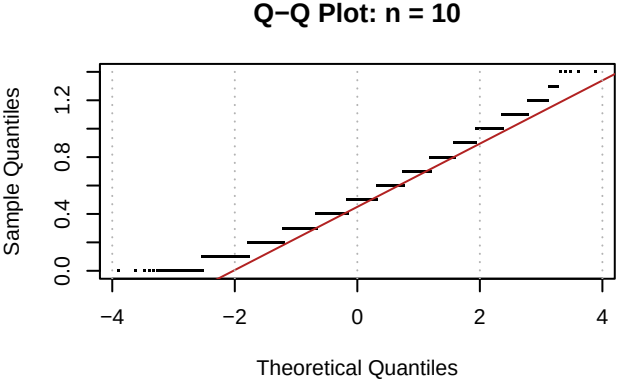

Sampling Distribution of Mean: n = 1
Density
0.0
1.0
2.0
3.0
0
1
2
3
4
Sample mean

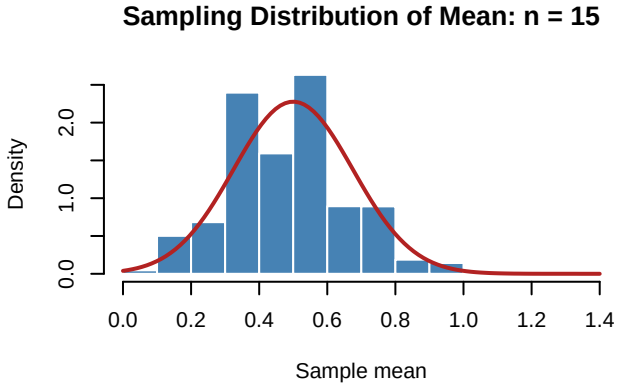

Q−Q Plot: n = 1
Sample Quantiles
0
1
2
3
4
−4
−2
0
2
4
Theoretical Quantiles

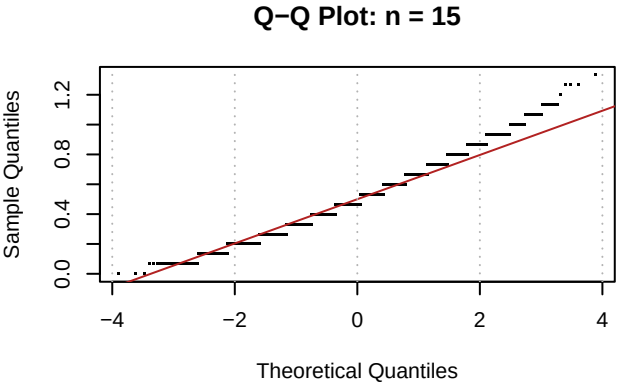

Sampling Distribution of Mean: n = 5
Density
0.0
0.4
0.8
1.2
0.0
0.5
1.0
1.5
2.0
Sample mean

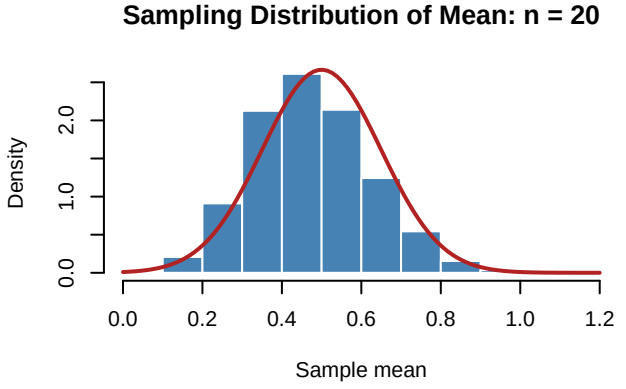

Q−Q Plot: n = 5
Sample Quantiles
0.0
1.0
2.0
−4
−2
0
2
4
Theoretical Quantiles

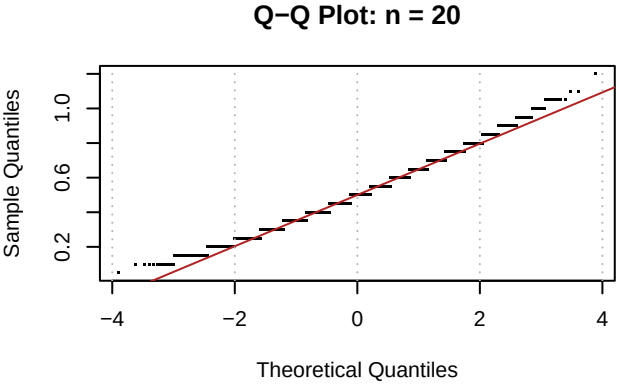

Sampling Distribution of Mean: n = 10
Density
0.0
0.5
1.0
1.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Sample mean

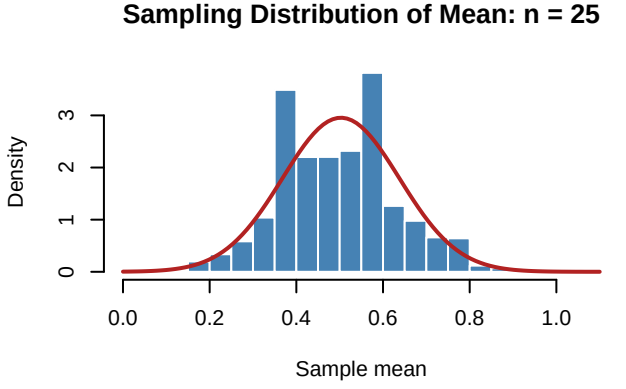

Q−Q Plot: n = 10
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

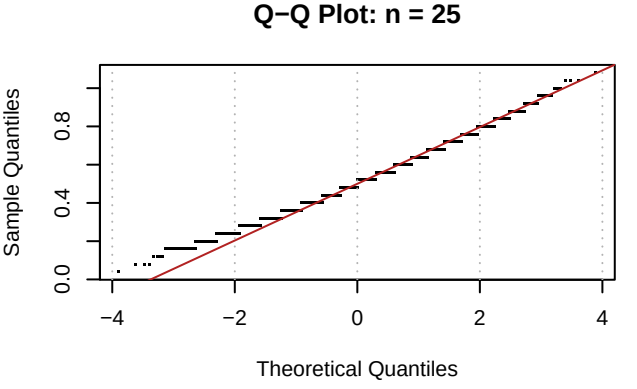

Sampling Distribution of Mean: n = 15
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Sample mean

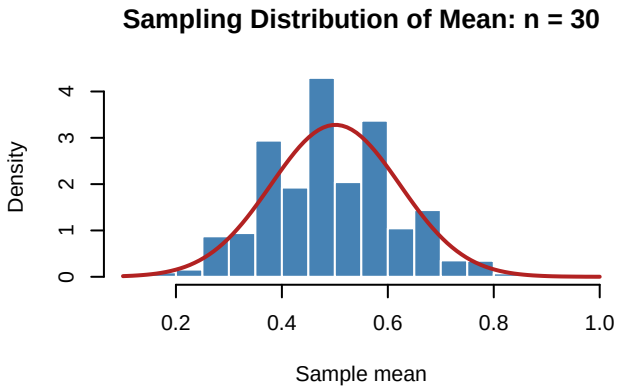

Q−Q Plot: n = 15
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

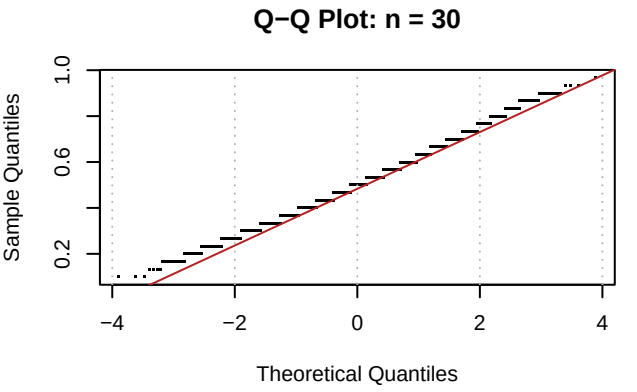

Sampling Distribution of Mean: n = 20
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

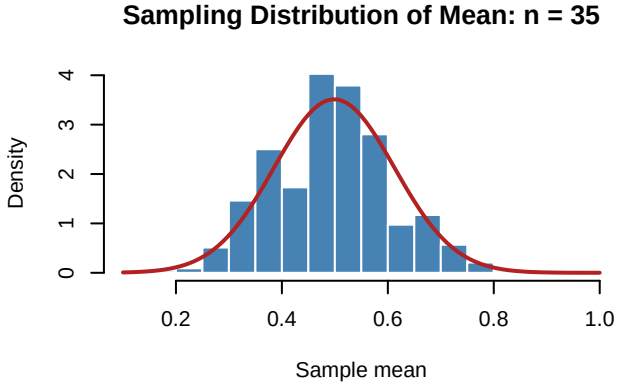

Q−Q Plot: n = 20
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

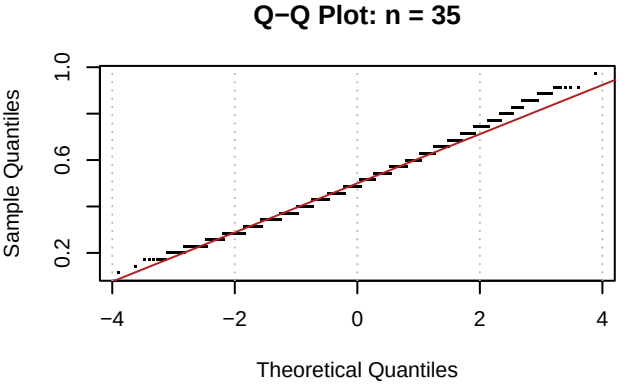

Sampling Distribution of Mean: n = 25
Density
0
1
2
3
0.0
0.2
0.4
0.6
0.8
1.0
Sample mean

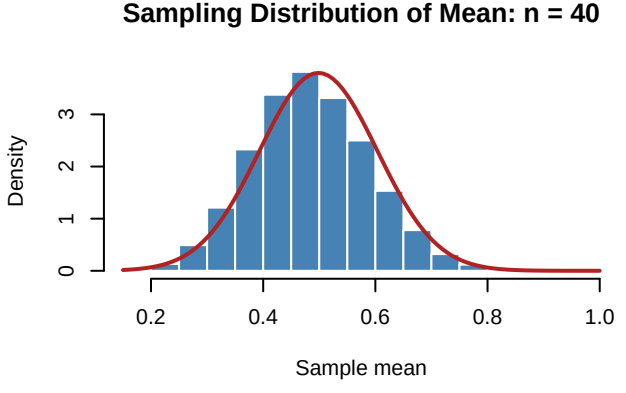

Q−Q Plot: n = 25
Sample Quantiles
0.0
0.4
0.8
−4
−2
0
2
4
Theoretical Quantiles

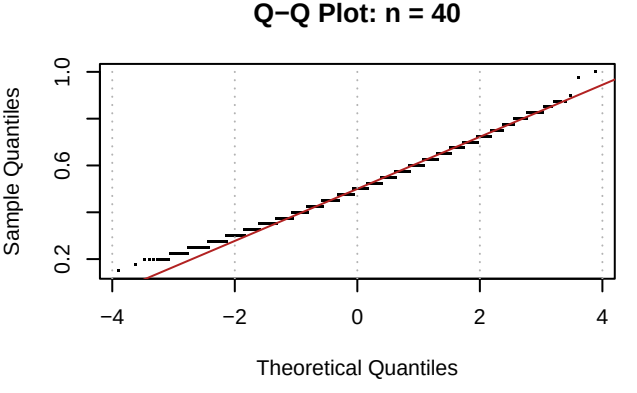

Sampling Distribution of Mean: n = 30
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

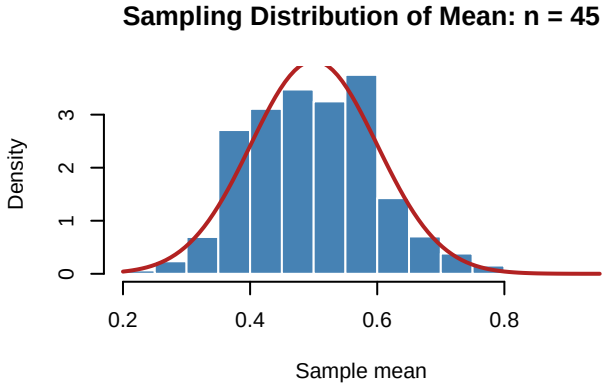

Q−Q Plot: n = 30
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

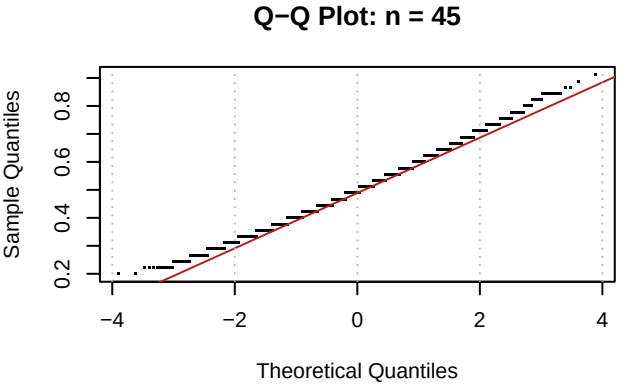

Sampling Distribution of Mean: n = 35
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

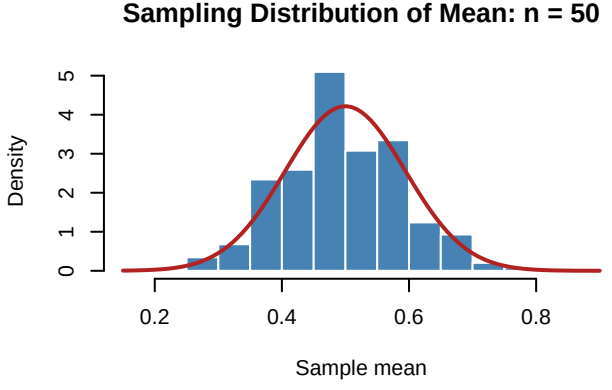

Q−Q Plot: n = 35
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

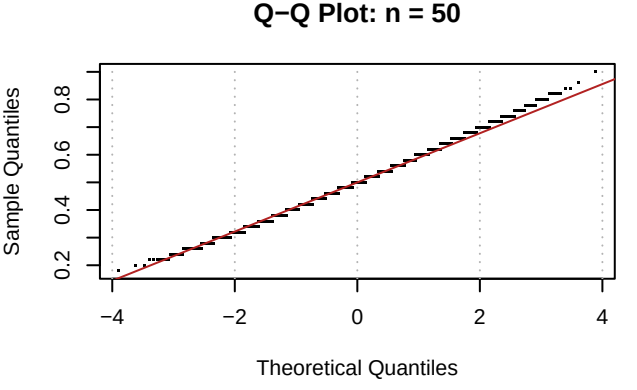

Sampling Distribution of Mean: n = 40
Density
0
1
2
3
0.2
0.4
0.6
0.8
1.0
Sample mean

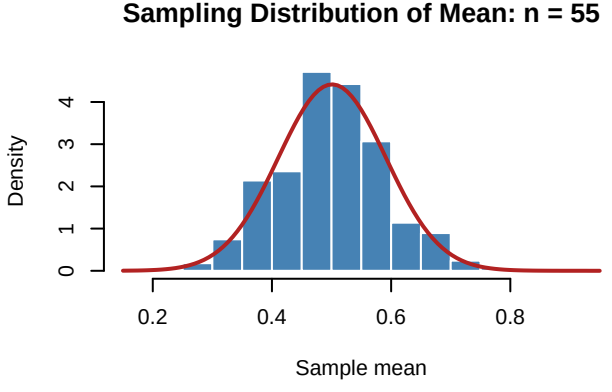

Q−Q Plot: n = 40
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

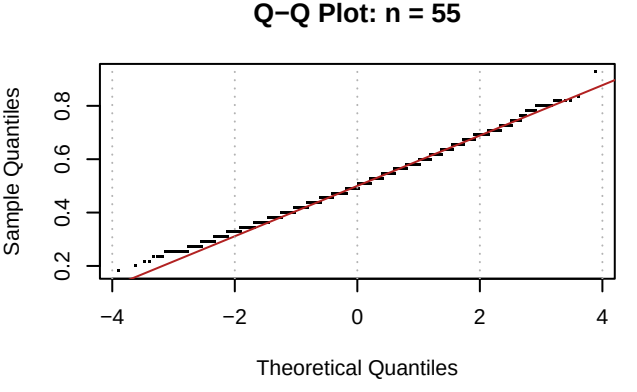

Sampling Distribution of Mean: n = 45
Density
0
1
2
3
0.2
0.4
0.6
0.8
Sample mean

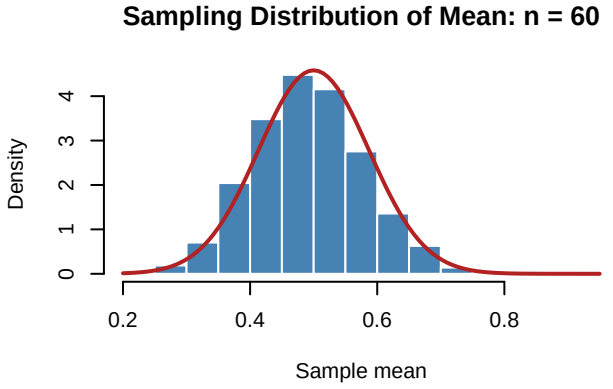

Q−Q Plot: n = 45
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

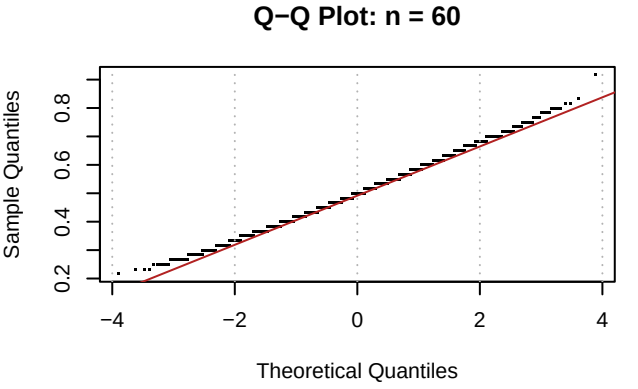

Sampling Distribution of Mean: n = 50
Density
0
1
2
3
4
5
0.2
0.4
0.6
0.8
Sample mean

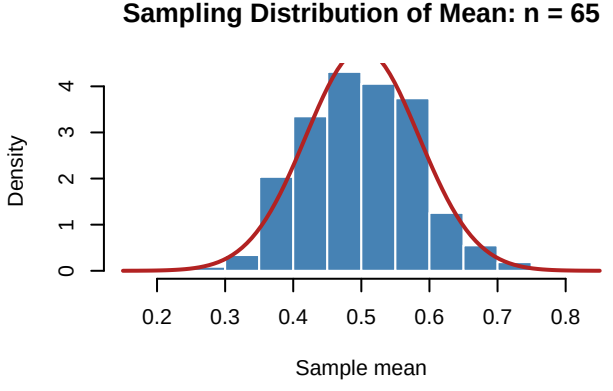

Q−Q Plot: n = 50
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

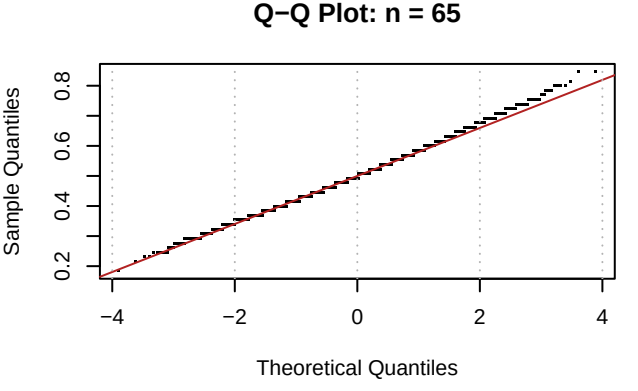

Sampling Distribution of Mean: n = 55
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
Sample mean

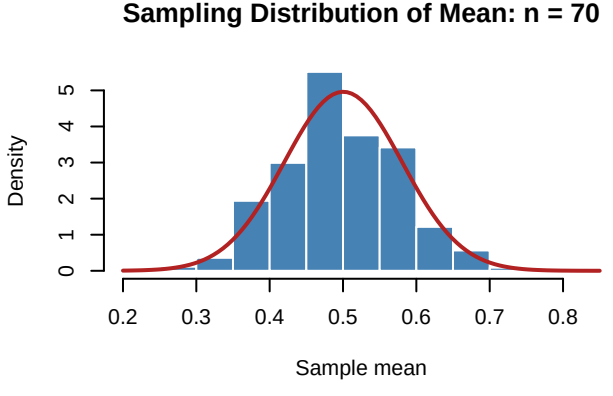

Q−Q Plot: n = 55
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

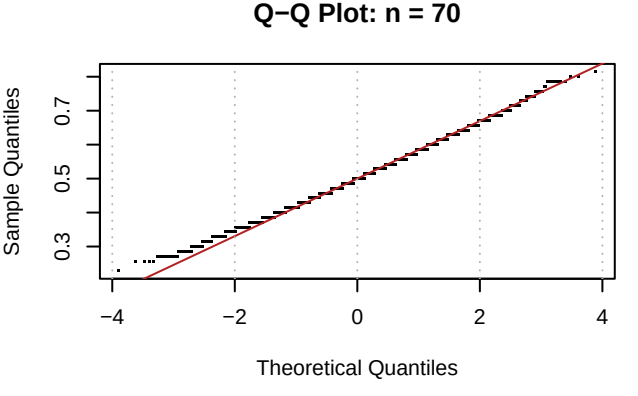

Sampling Distribution of Mean: n = 60
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
Sample mean

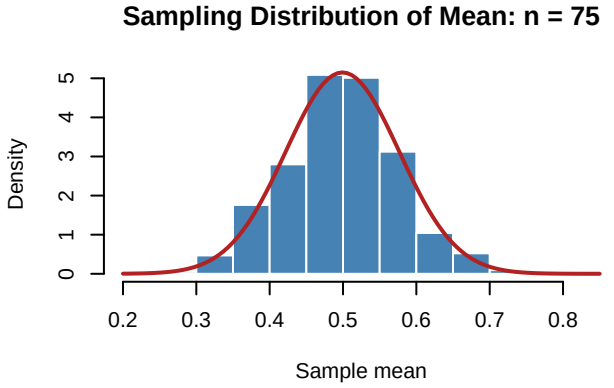

Q−Q Plot: n = 60
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

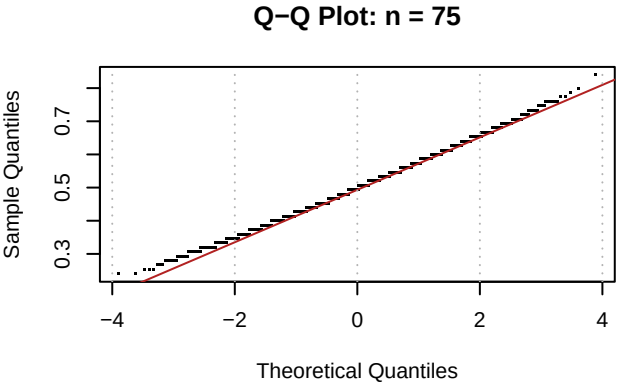

Sampling Distribution of Mean: n = 65
Density
0
1
2
3
4
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

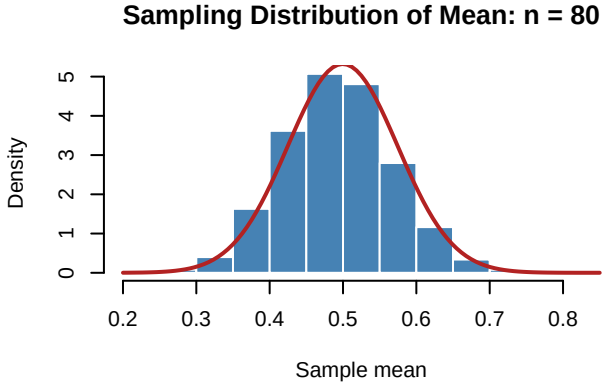

Q−Q Plot: n = 65
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

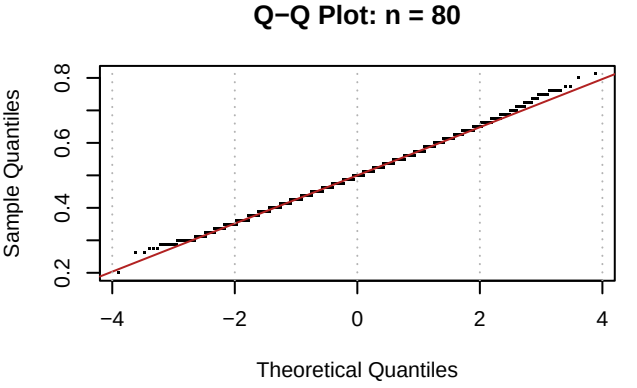

Sampling Distribution of Mean: n = 70
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

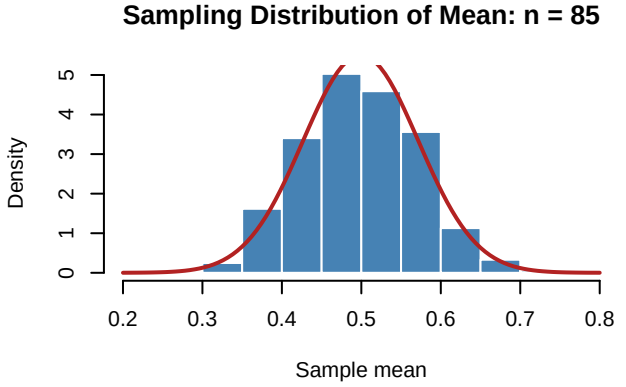

Q−Q Plot: n = 70
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

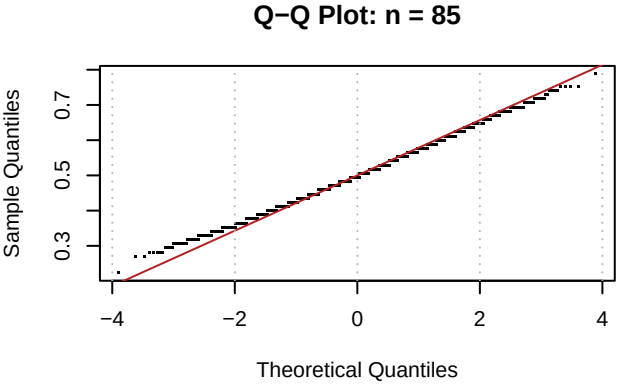

Sampling Distribution of Mean: n = 75
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

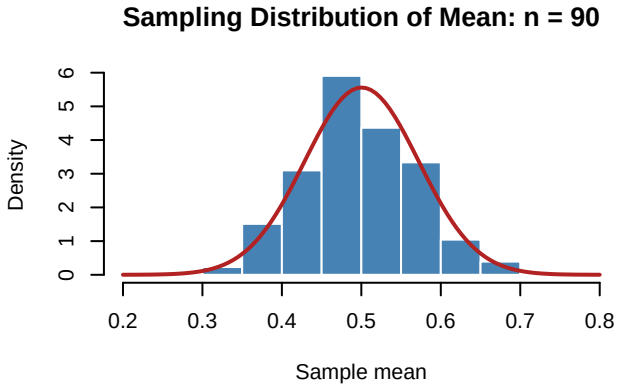

Q−Q Plot: n = 75
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

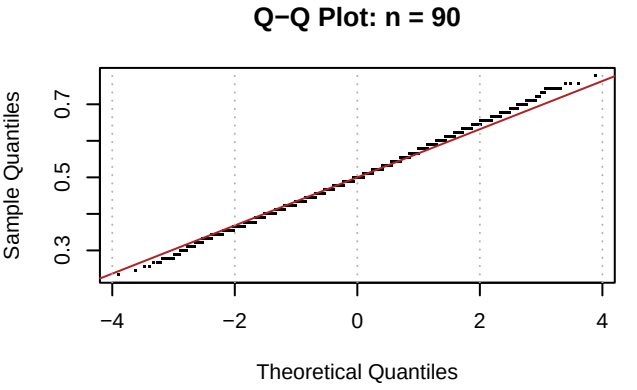

Sampling Distribution of Mean: n = 80
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

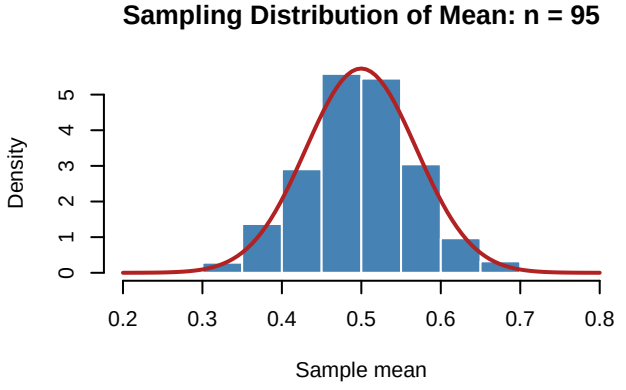

Q−Q Plot: n = 80
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

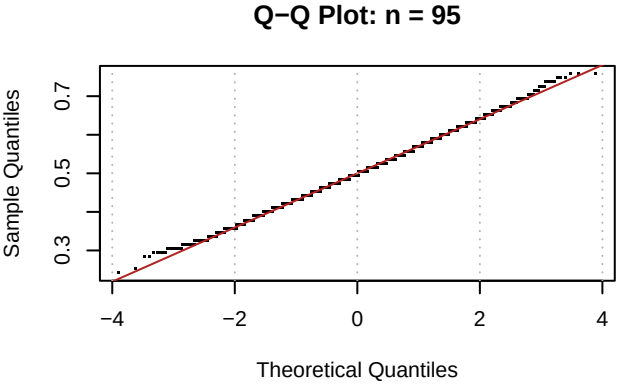

Sampling Distribution of Mean: n = 85
Density
0
1
2
3
4
5
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

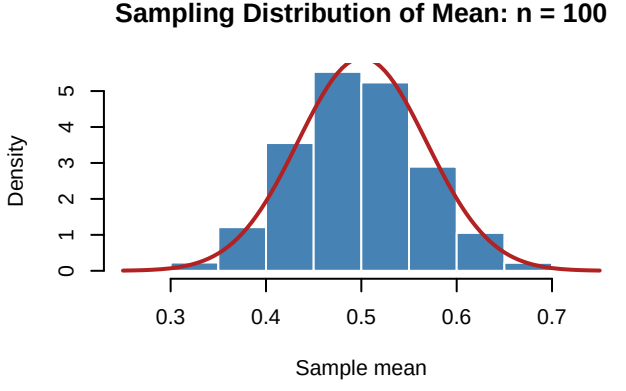

Q−Q Plot: n = 85
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

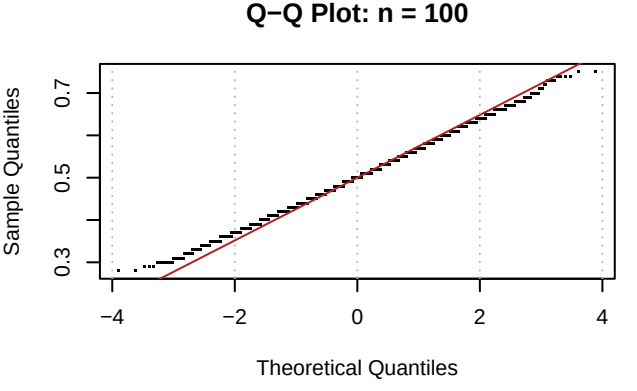

Sampling Distribution of Mean: n = 90
Density
0 1 2 3 4 5 6
0.2 0.3 0.4 0.5 0.6 0.7 0.8
Sample mean

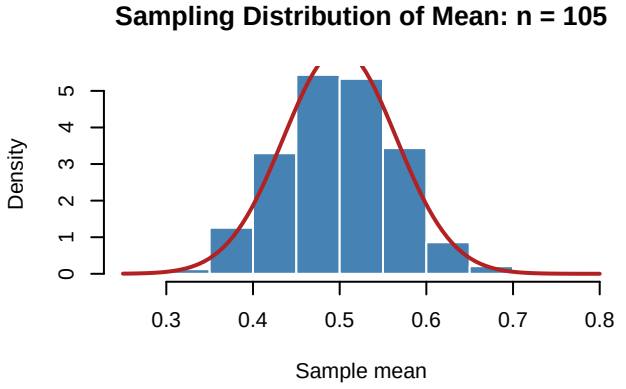

Q−Q Plot: n = 90
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

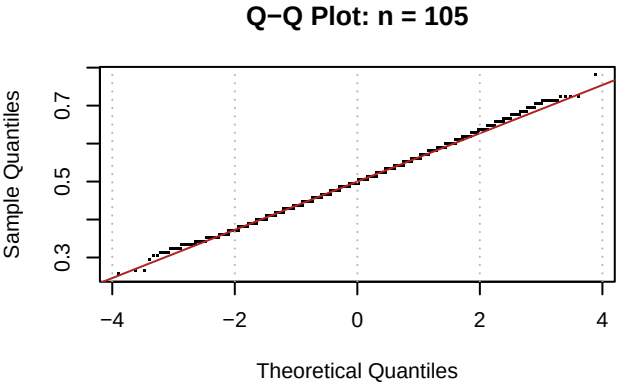

Sampling Distribution of Mean: n = 95
Density
0 1 2 3 4 5
0.2 0.3 0.4 0.5 0.6 0.7 0.8
Sample mean

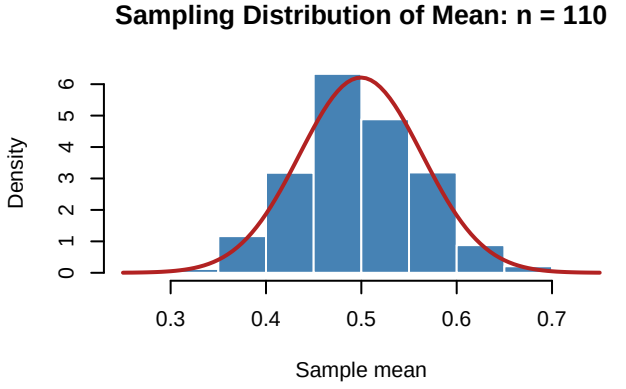

Q−Q Plot: n = 95
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

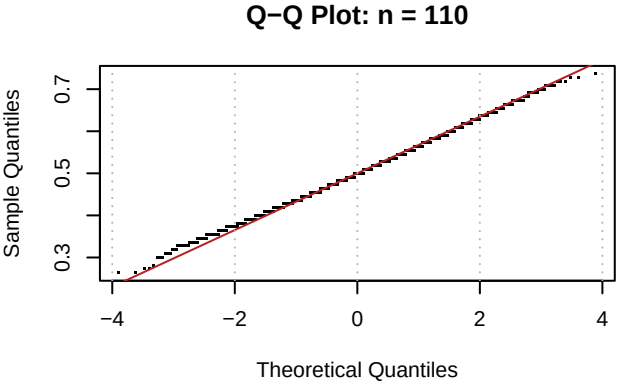

Sampling Distribution of Mean: n = 100
Density
0 1 2 3 4 5
0.3 0.4 0.5 0.6 0.7
Sample mean

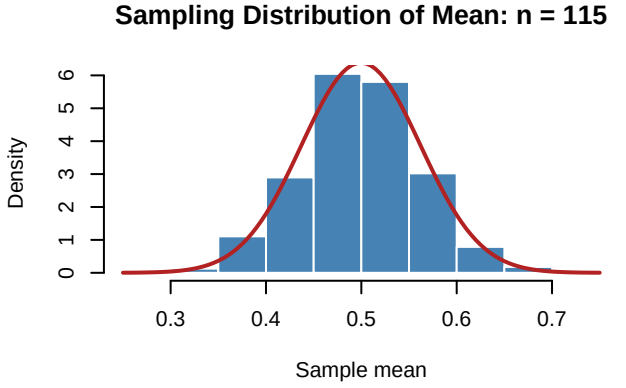

Q−Q Plot: n = 100
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

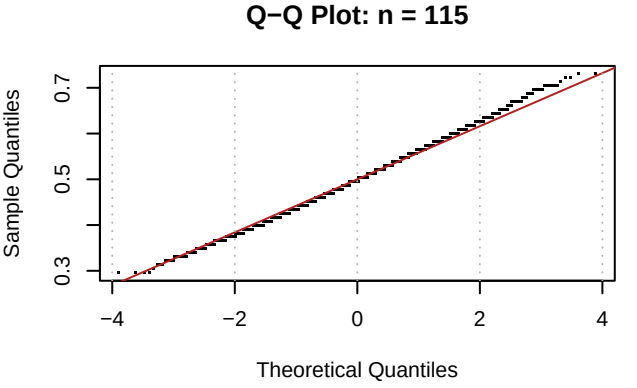

Sampling Distribution of Mean: n = 105
Density
0
1
2
3
4
5
0.3
0.4
0.5
0.6
0.7
0.8
Sample mean

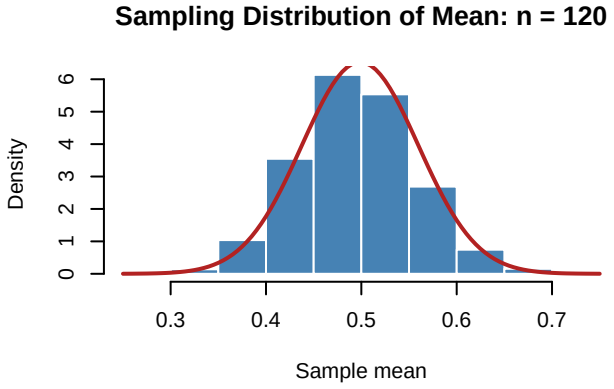

Q−Q Plot: n = 105
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

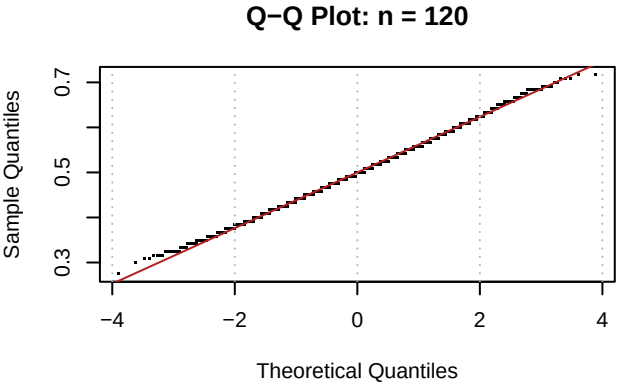

Sampling Distribution of Mean: n = 110
Density
0
1
2
3
4
5
6
0.3
0.4
0.5
0.6
0.7
Sample mean

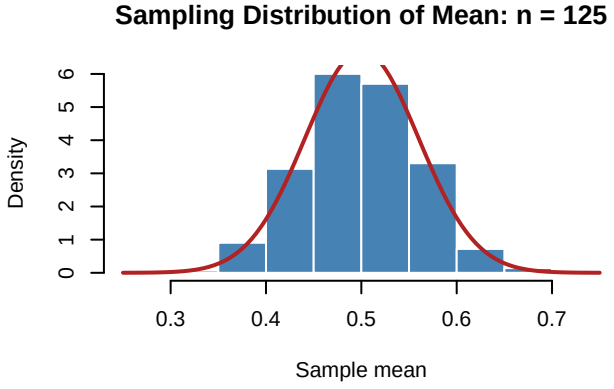

Q−Q Plot: n = 110
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

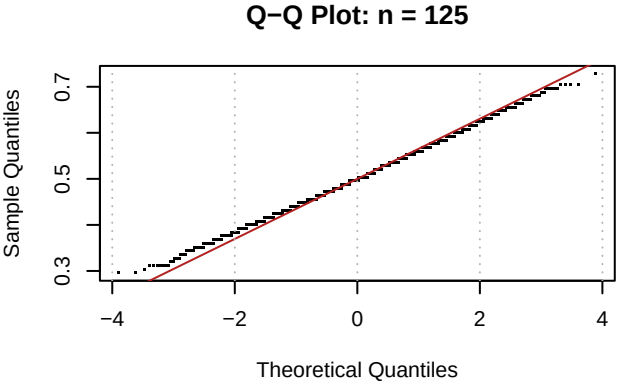

Sampling Distribution of Mean: n = 115
Density
0
1
2
3
4
5
6
0.3
0.4
0.5
0.6
0.7
Sample mean

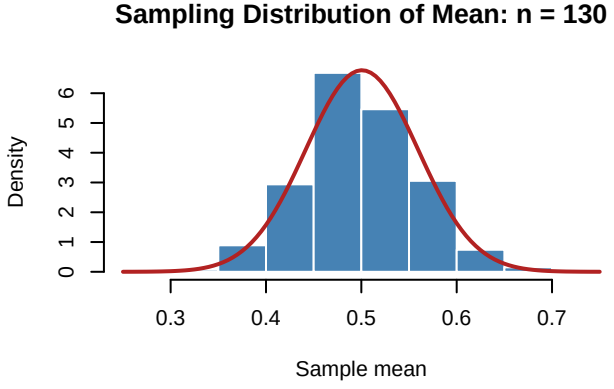

Q−Q Plot: n = 115
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

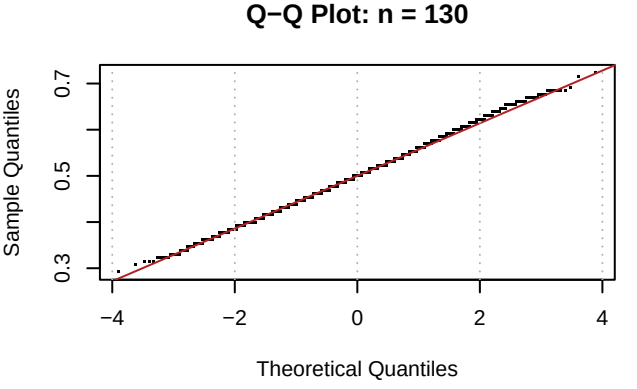

Sampling Distribution of Mean: n = 120
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

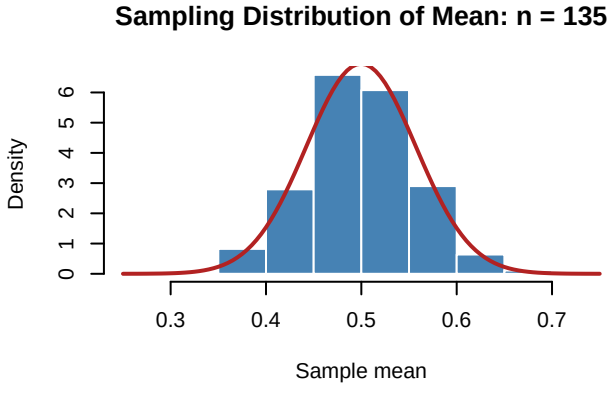

Q−Q Plot: n = 120
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

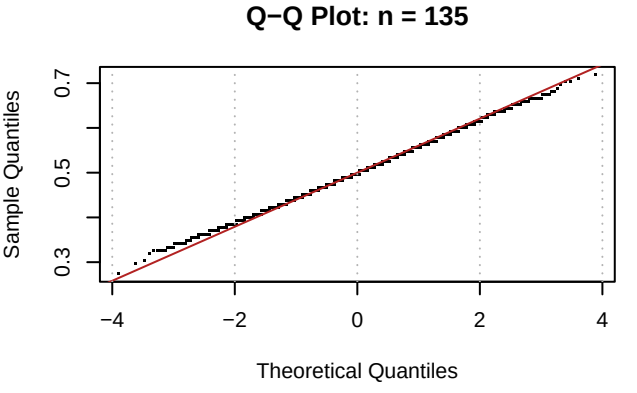

Sampling Distribution of Mean: n = 125
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

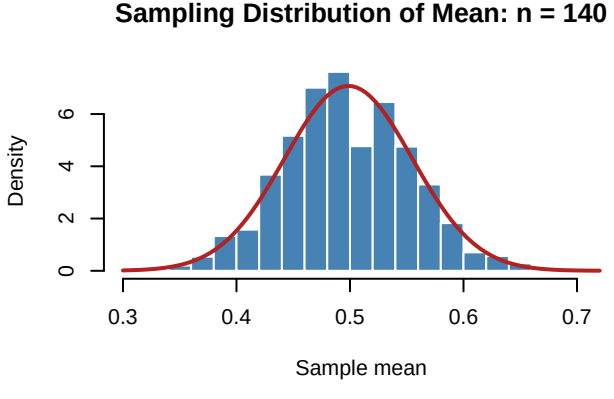

Q−Q Plot: n = 125
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

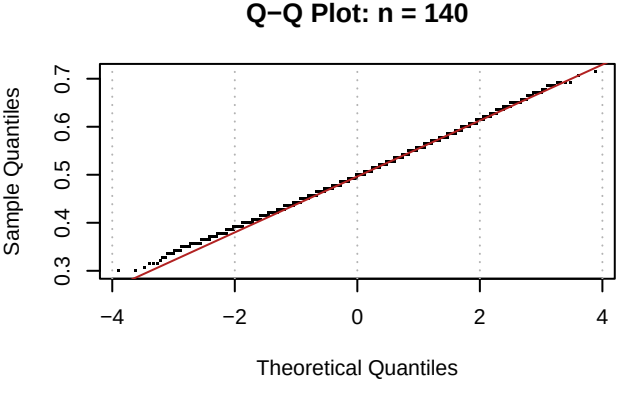

Sampling Distribution of Mean: n = 130
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

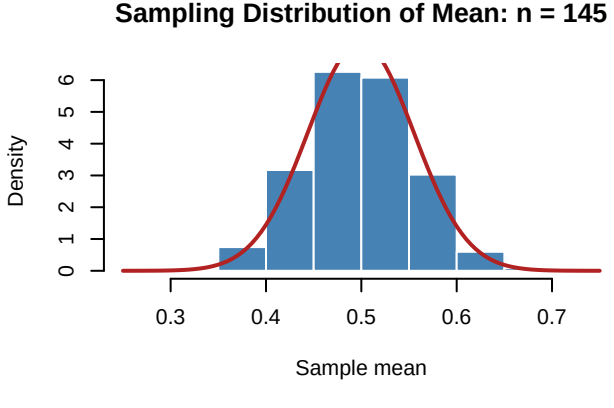

Q−Q Plot: n = 130
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

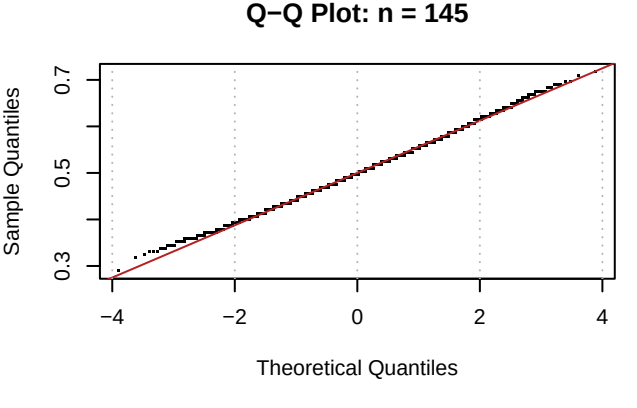

Sampling Distribution of Mean: n = 135
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

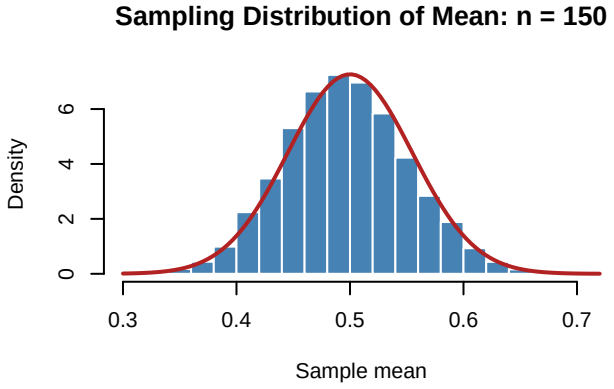

Q−Q Plot: n = 135
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

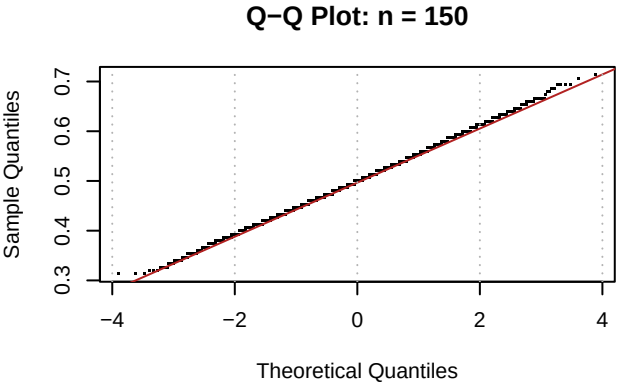

Sampling Distribution of Mean: n = 140
Density
0 2 4 6
0.3 0.4 0.5 0.6 0.7
Sample mean

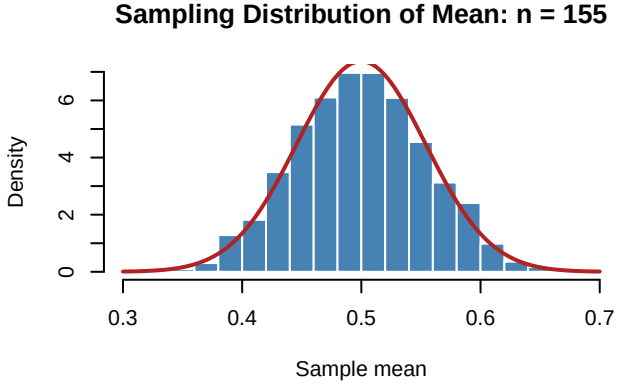

Q−Q Plot: n = 140
Sample Quantiles
0.3 0.4 0.5 0.6 0.7
−4 −2 0 2 4
Theoretical Quantiles

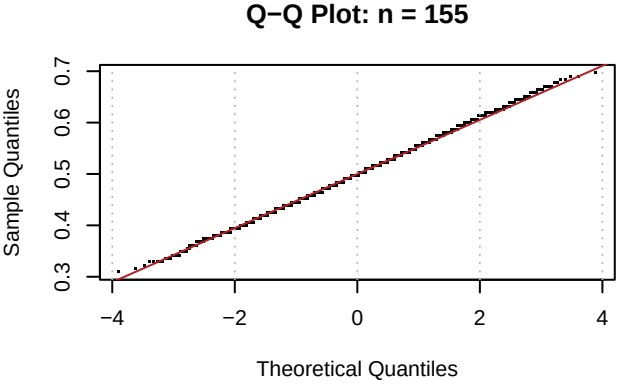

Sampling Distribution of Mean: n = 145
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

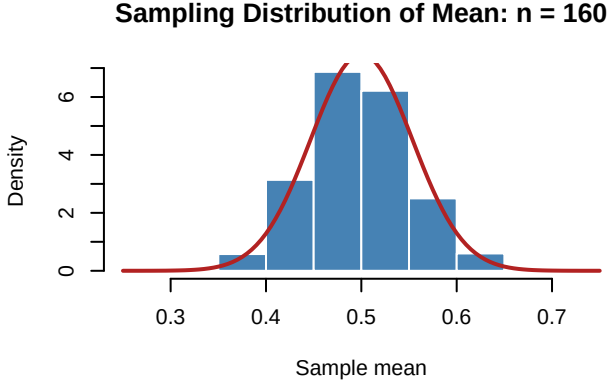

Q−Q Plot: n = 145
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

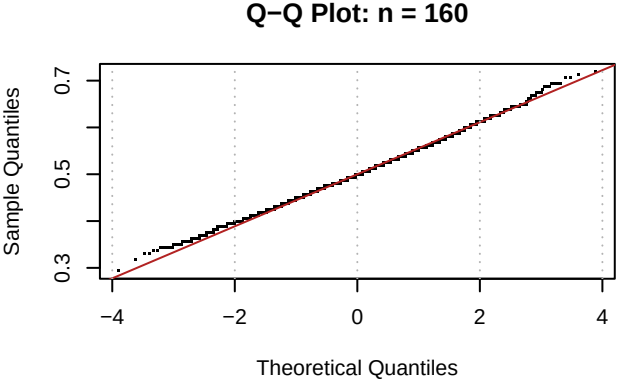

Sampling Distribution of Mean: n = 150
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

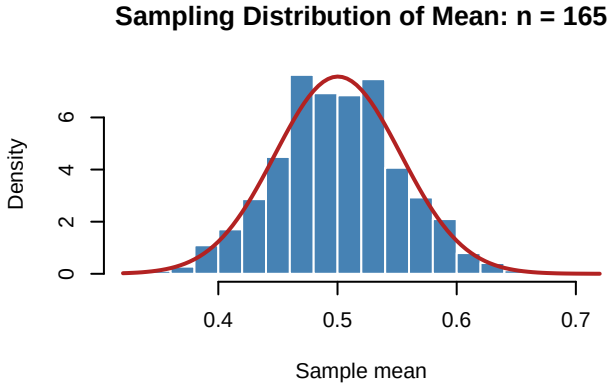

Q−Q Plot: n = 150
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

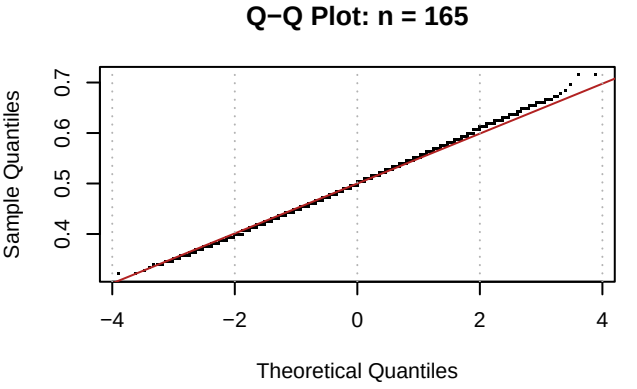

Sampling Distribution of Mean: n = 155
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

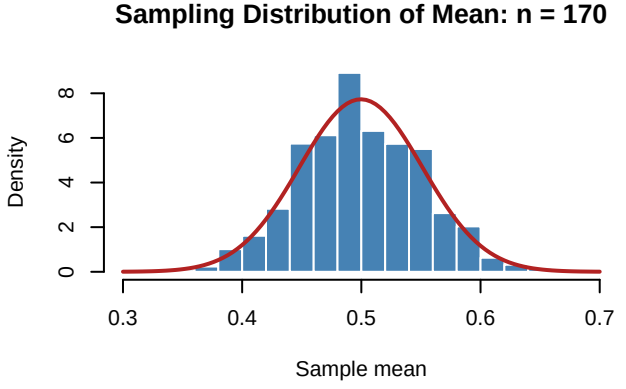

Q−Q Plot: n = 155
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

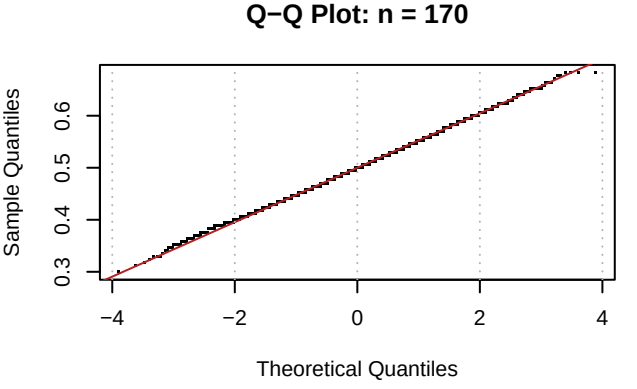

Sampling Distribution of Mean: n = 160
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

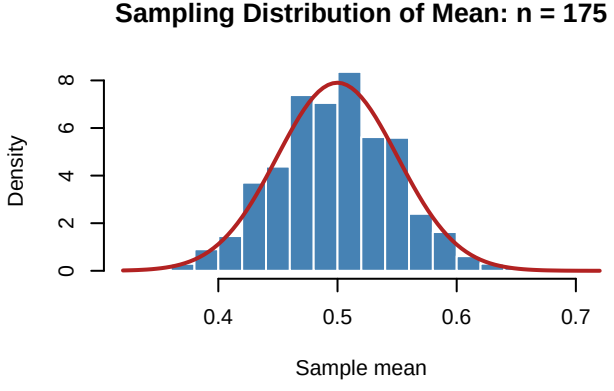

Q−Q Plot: n = 160
Sample Quantiles
0.3
0.5
0.7
−4
−2
0
2
4
Theoretical Quantiles

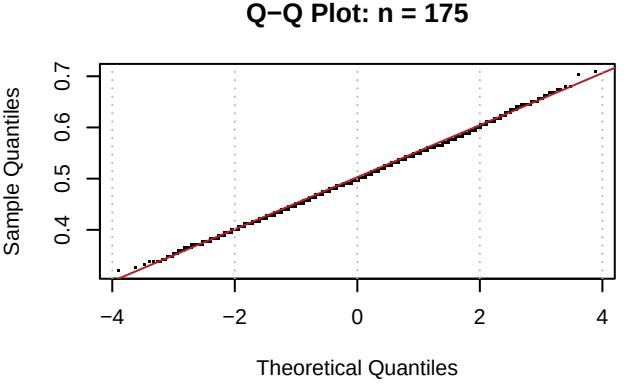

Sampling Distribution of Mean: n = 165
Density
0
2
4
6
0.4
0.5
0.6
0.7
Sample mean

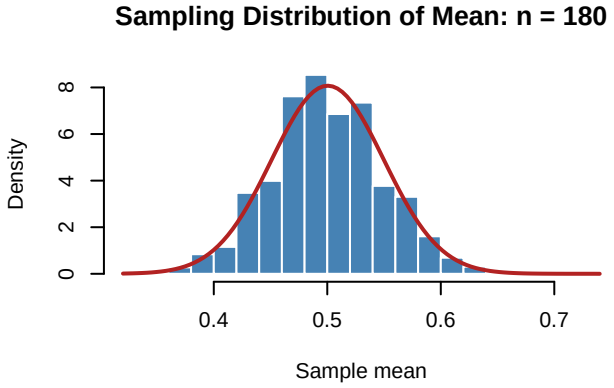

Q−Q Plot: n = 165
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

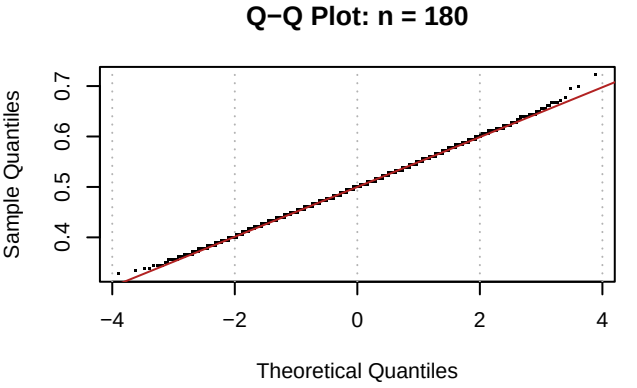

Sampling Distribution of Mean: n = 170
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
0.7
Sample mean

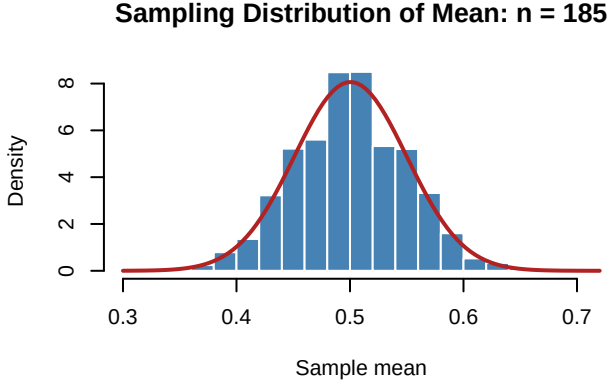

Q−Q Plot: n = 170
Sample Quantiles
0.3
0.4
0.5
0.6
−4
−2
0
2
4
Theoretical Quantiles

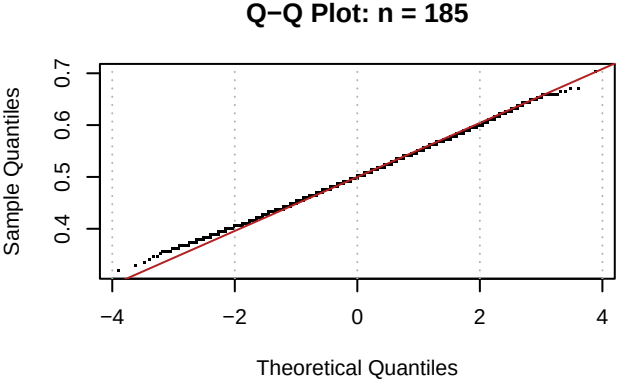

Sampling Distribution of Mean: n = 175
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

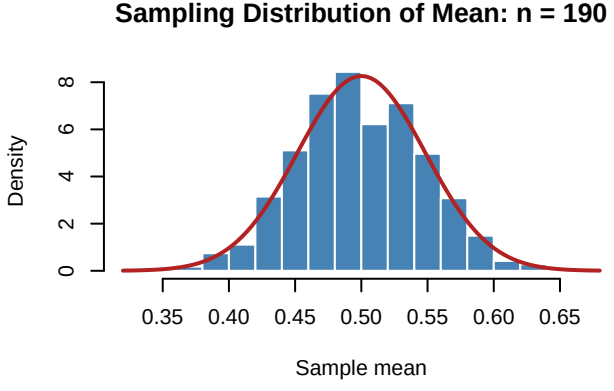

Q−Q Plot: n = 175
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

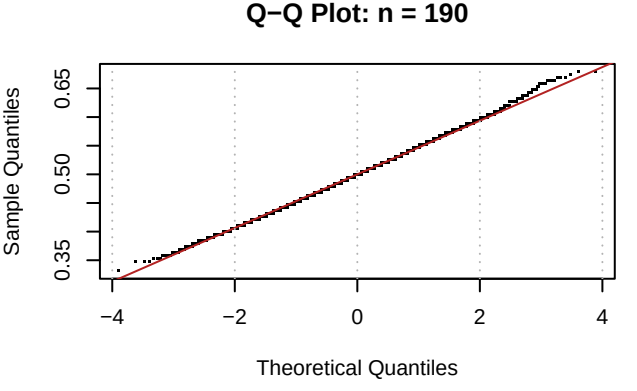

Sampling Distribution of Mean: n = 180
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

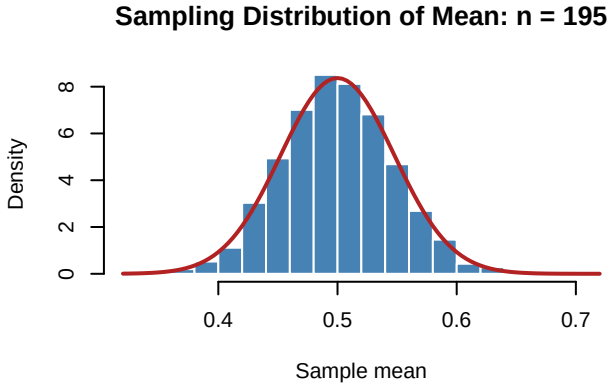

Q−Q Plot: n = 180
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

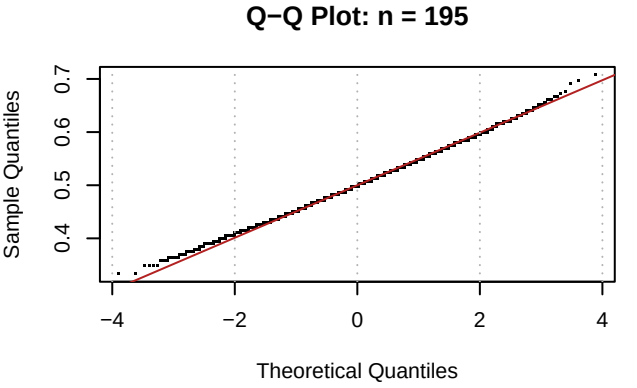

Sampling Distribution of Mean: n = 185
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
0.7
Sample mean

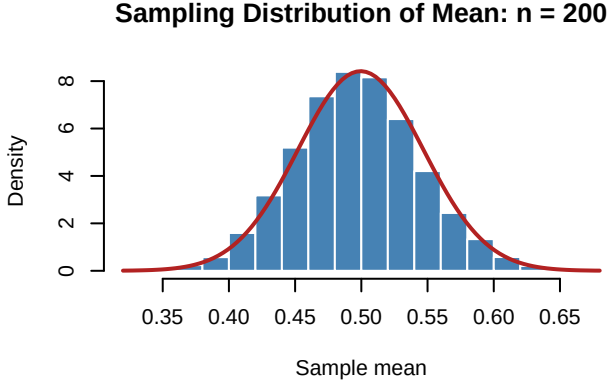

Q−Q Plot: n = 185
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

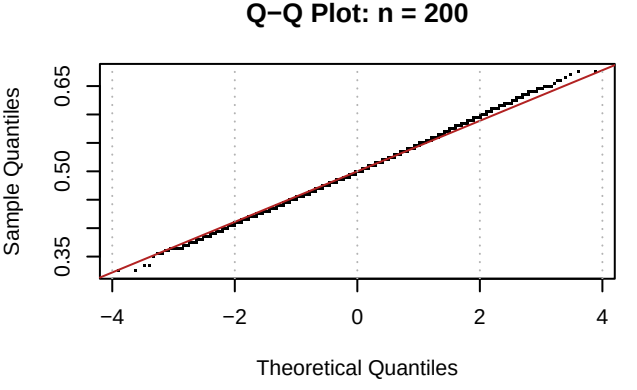

Sampling Distribution of Mean: n = 190
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

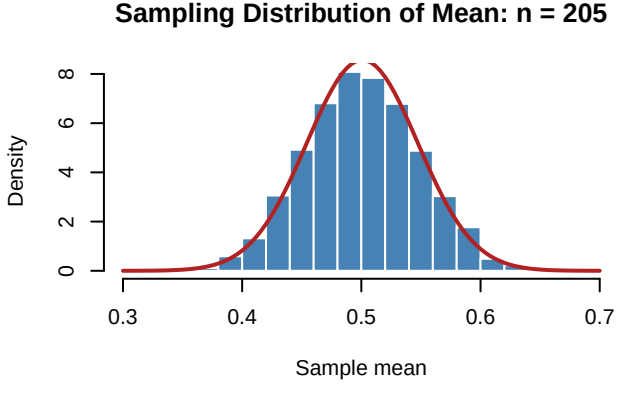

Q−Q Plot: n = 190
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

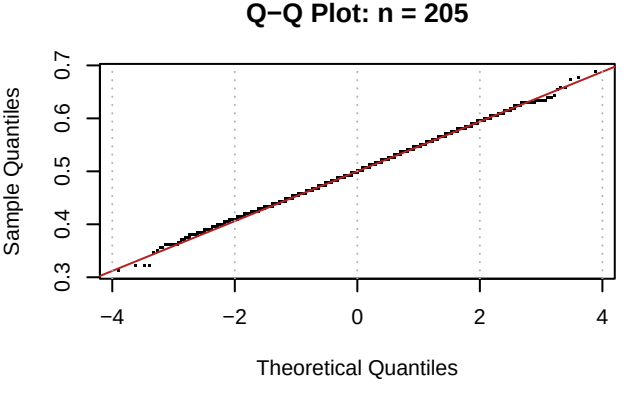

Sampling Distribution of Mean: n = 195
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

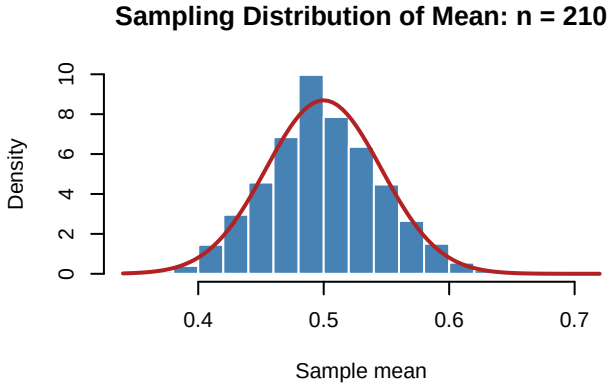

Q−Q Plot: n = 195
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

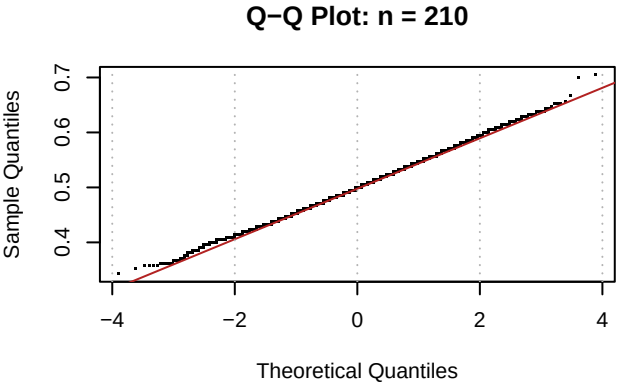

Sampling Distribution of Mean: n = 200
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

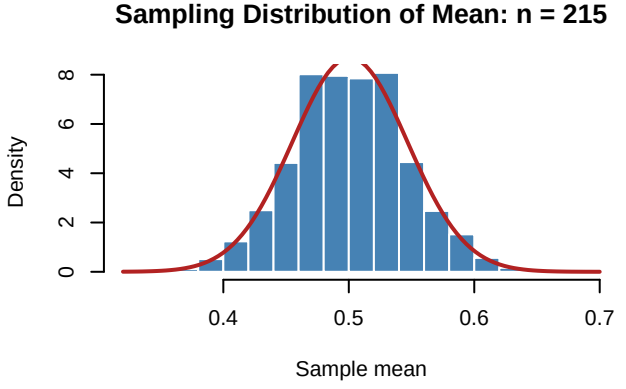

Q−Q Plot: n = 200
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

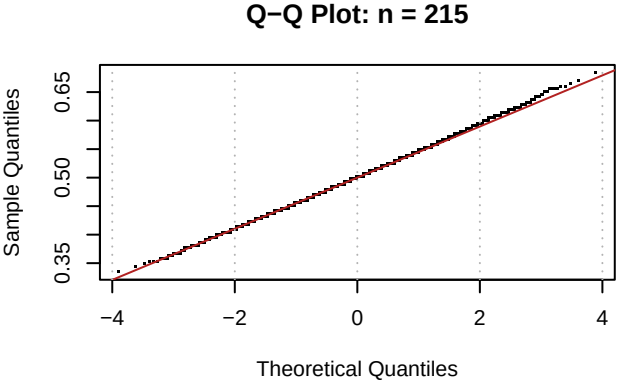

Sampling Distribution of Mean: n = 205
Density
0
2
4
6
8
0.3
0.4
0.5
0.6
0.7
Sample mean

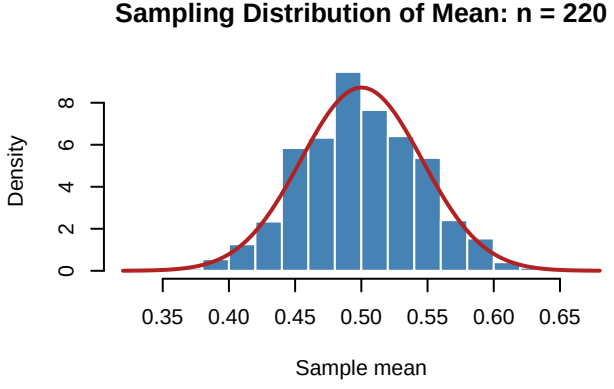

Q−Q Plot: n = 205
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

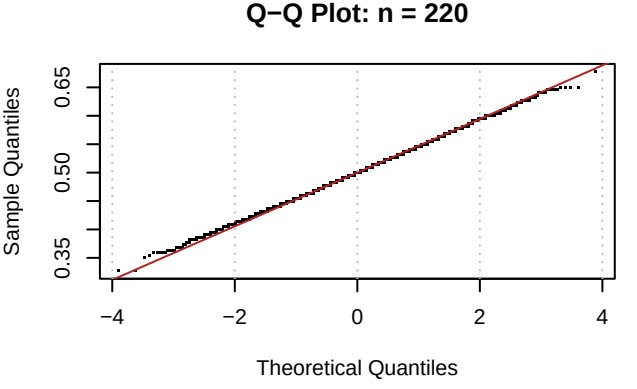

Sampling Distribution of Mean: n = 210
Density
0
2
4
6
8
10
0.4
0.5
0.6
0.7
Sample mean

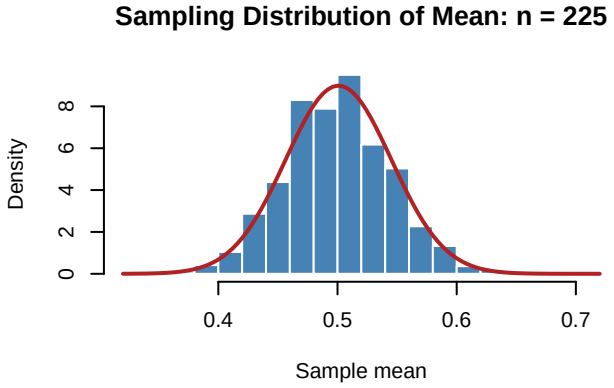

Q−Q Plot: n = 210
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

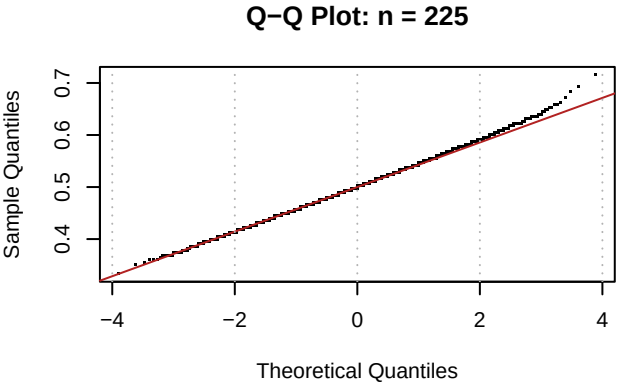

Sampling Distribution of Mean: n = 215
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

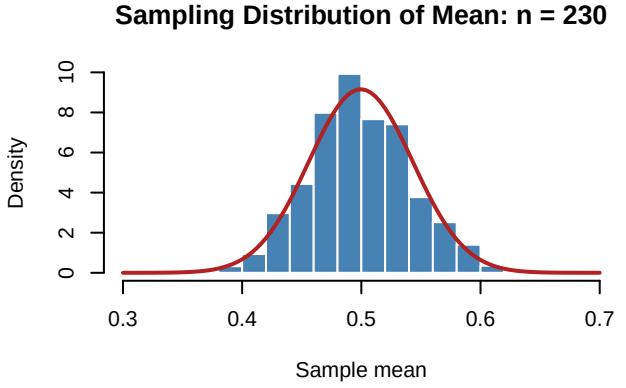

Q−Q Plot: n = 215
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

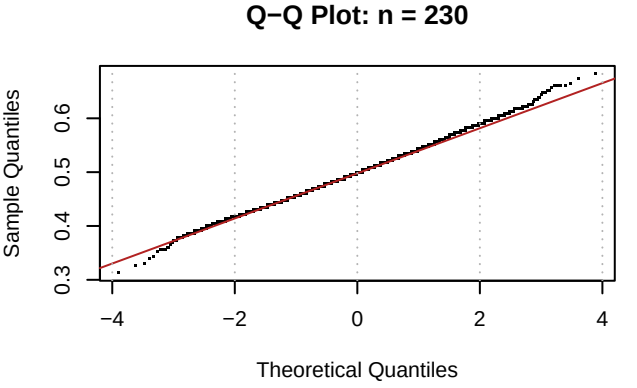

Sampling Distribution of Mean: n = 220
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

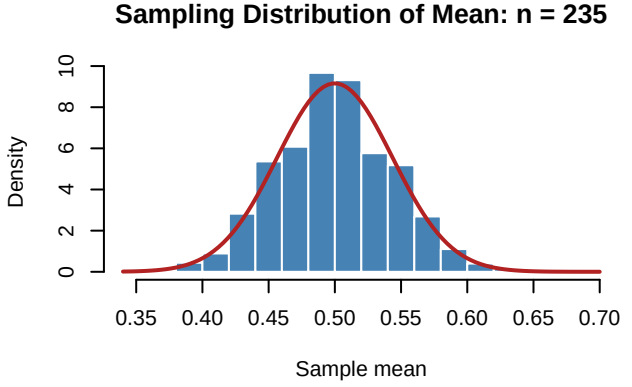

Q−Q Plot: n = 220
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

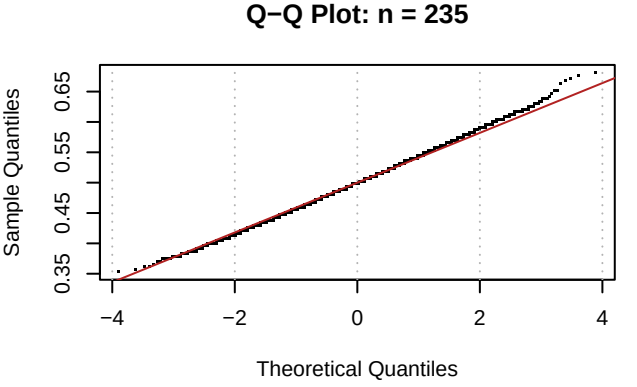

Sampling Distribution of Mean: n = 225
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

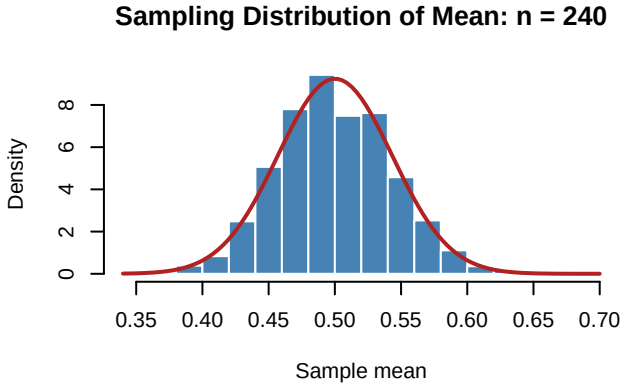

Q−Q Plot: n = 225
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

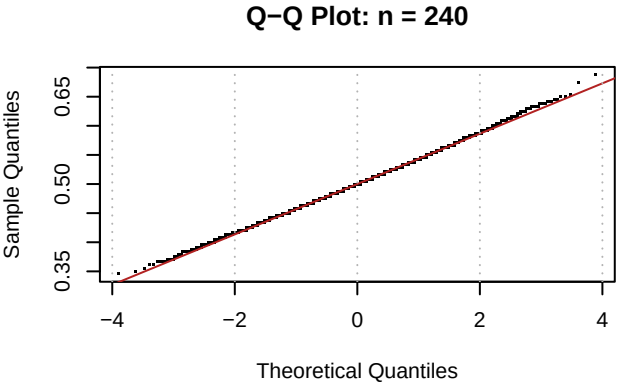

Sampling Distribution of Mean: n = 230
Density
0
2
4
6
8
10
0.3
0.4
0.5
0.6
0.7
Sample mean

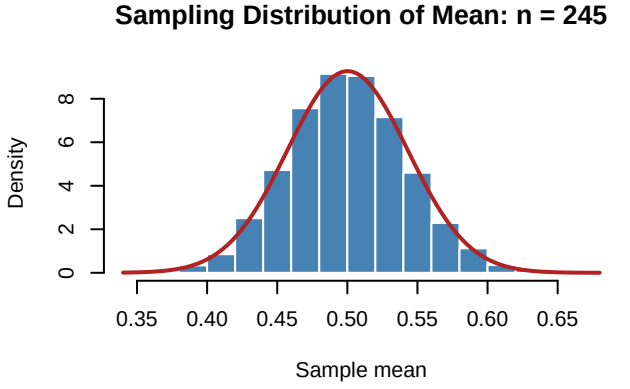

Q−Q Plot: n = 230
Sample Quantiles
0.3
0.4
0.5
0.6
−4
−2
0
2
4
Theoretical Quantiles

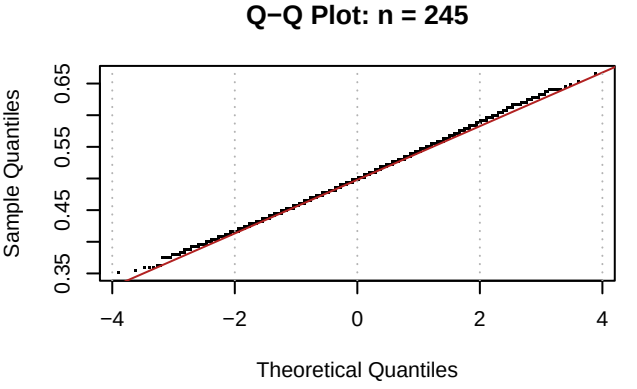

Sampling Distribution of Mean: n = 235
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
Sample mean

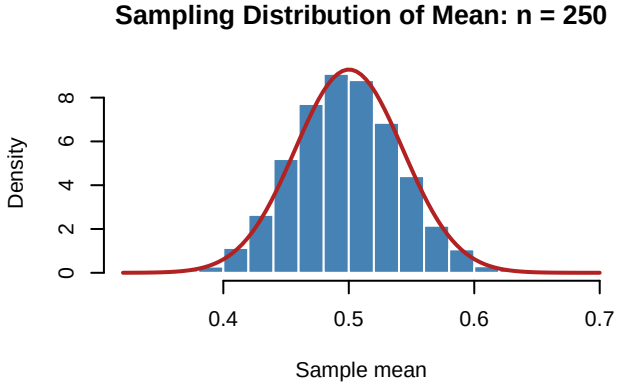

Q−Q Plot: n = 235
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

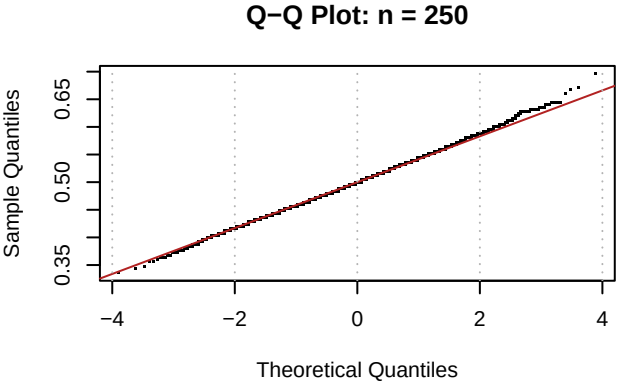

Sampling Distribution of Mean: n = 240
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
Sample mean

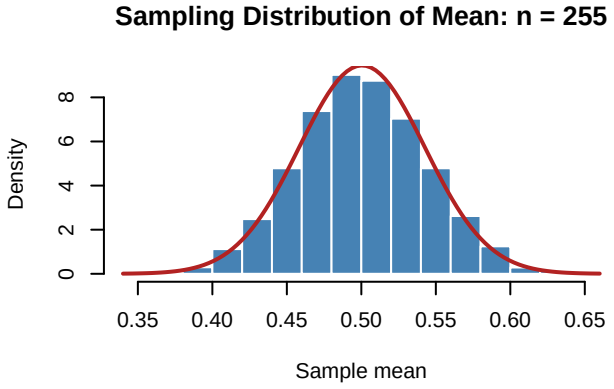

Q−Q Plot: n = 240
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

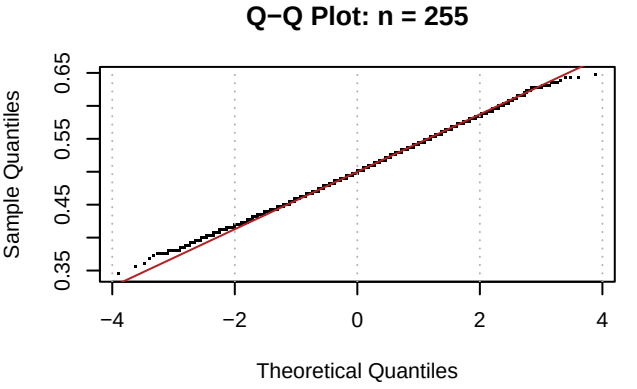

Sampling Distribution of Mean: n = 245
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

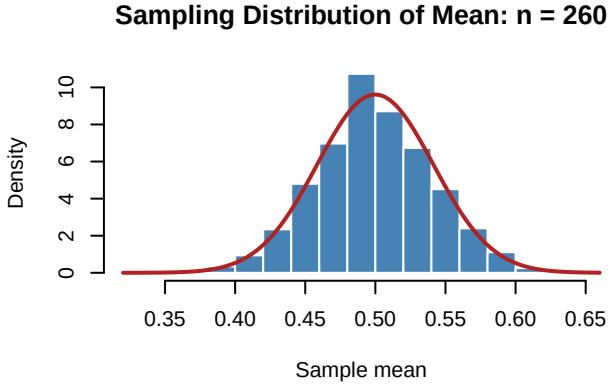

Q−Q Plot: n = 245
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

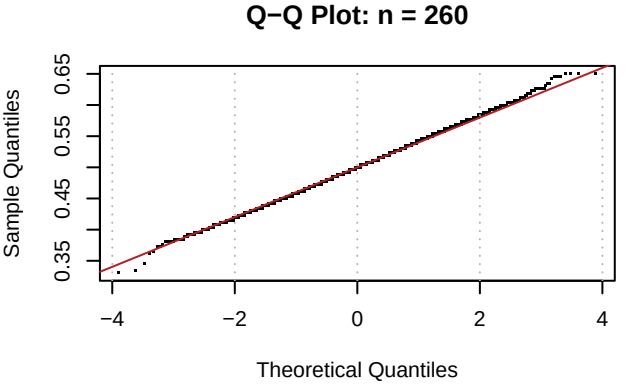

Sampling Distribution of Mean: n = 250
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

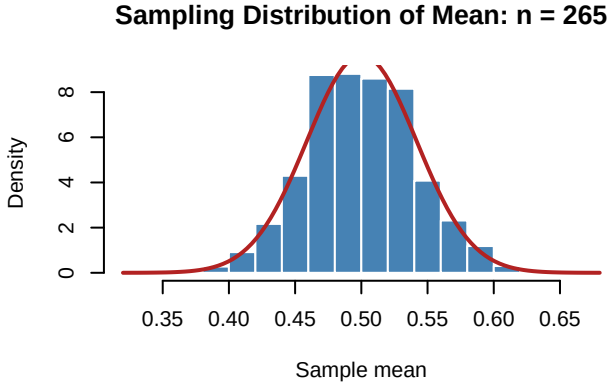

Q−Q Plot: n = 250
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

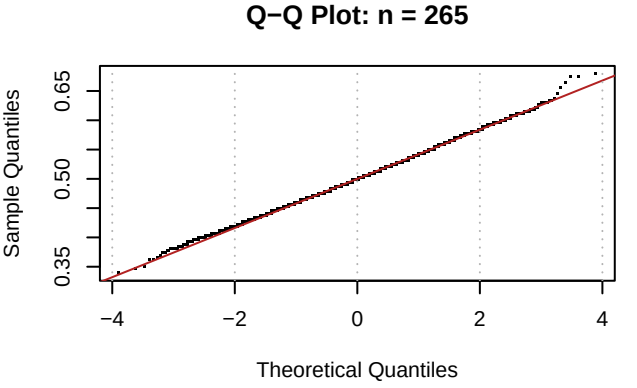

Sampling Distribution of Mean: n = 255
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

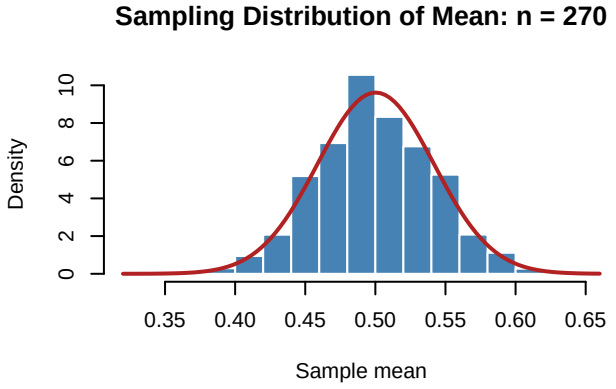

Q−Q Plot: n = 255
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

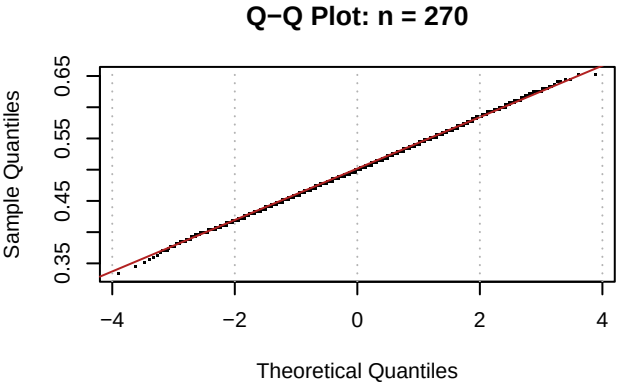

Sampling Distribution of Mean: n = 260
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

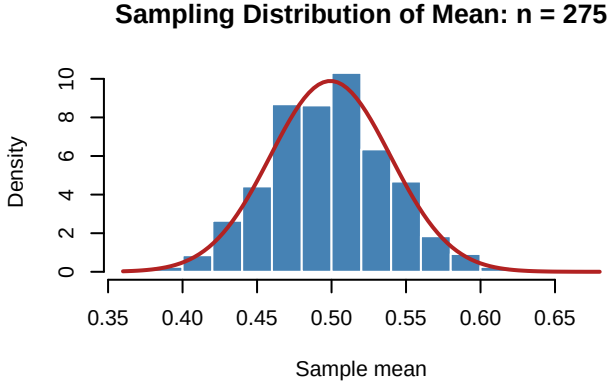

Q−Q Plot: n = 260
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

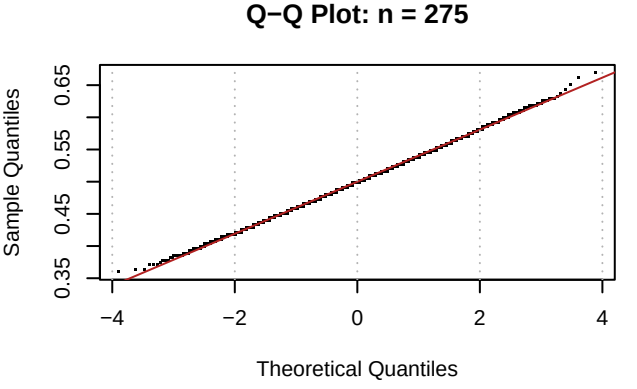

Sampling Distribution of Mean: n = 265
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

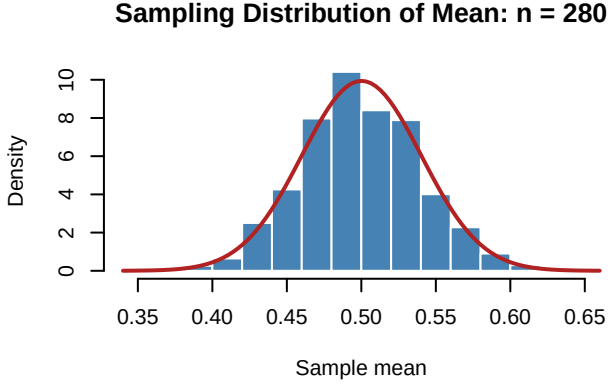

Q−Q Plot: n = 265
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

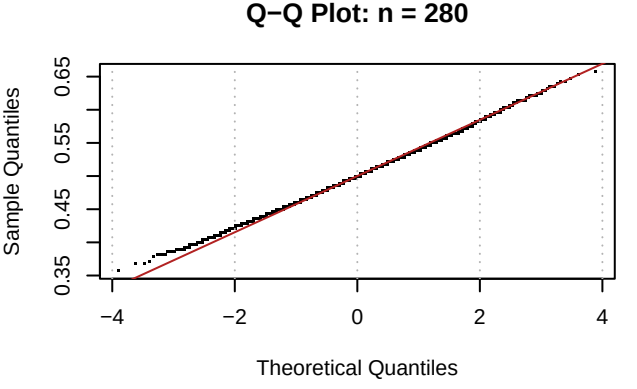

Sampling Distribution of Mean: n = 270
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

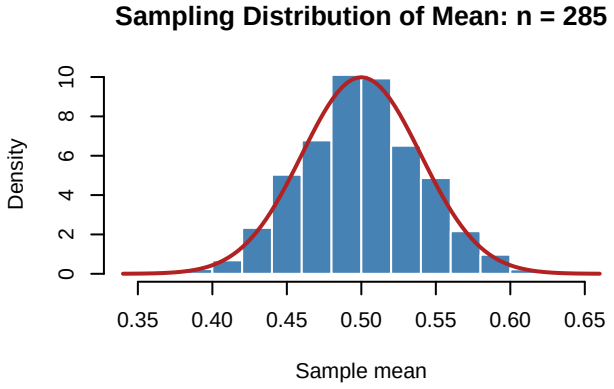

Q−Q Plot: n = 270
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

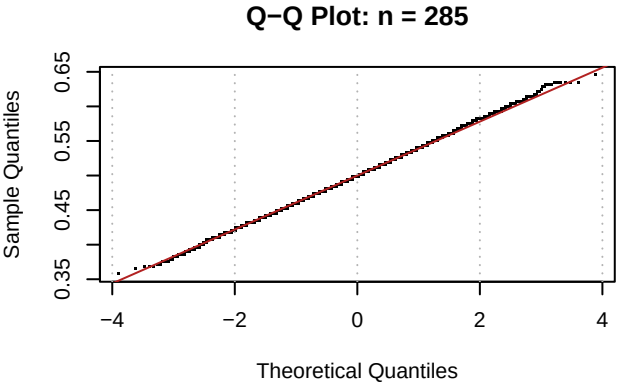

Sampling Distribution of Mean: n = 275
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

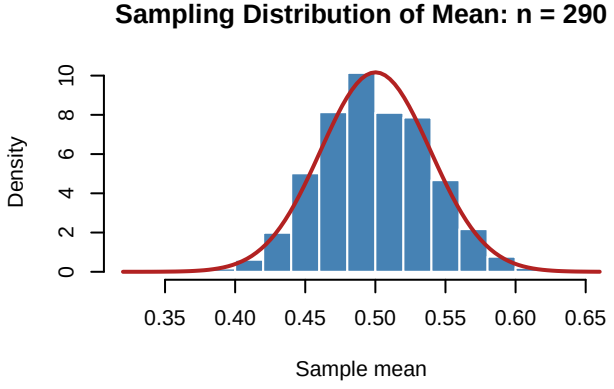

Q−Q Plot: n = 275
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

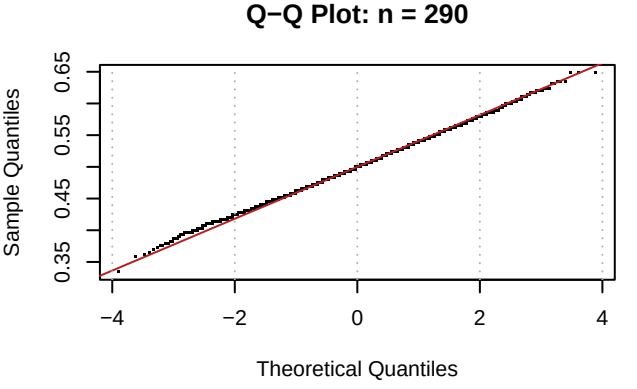

Sampling Distribution of Mean: n = 280
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

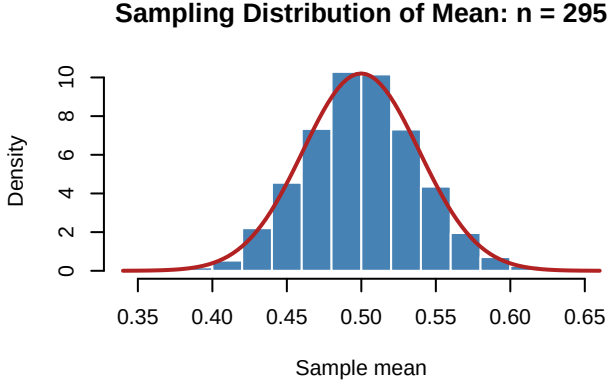

Q−Q Plot: n = 280
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

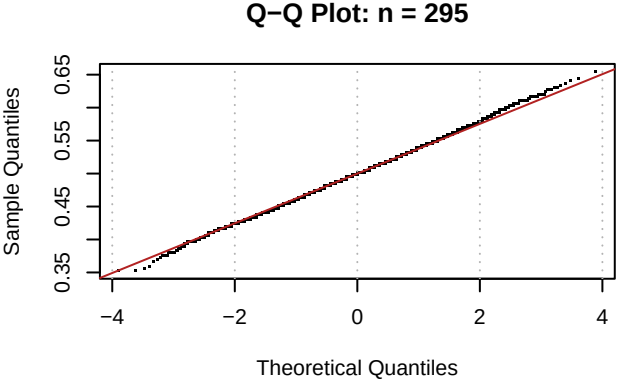

Sampling Distribution of Mean: n = 285
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

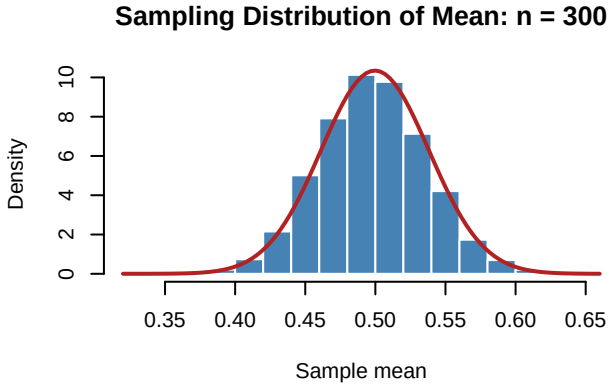

Q−Q Plot: n = 285
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

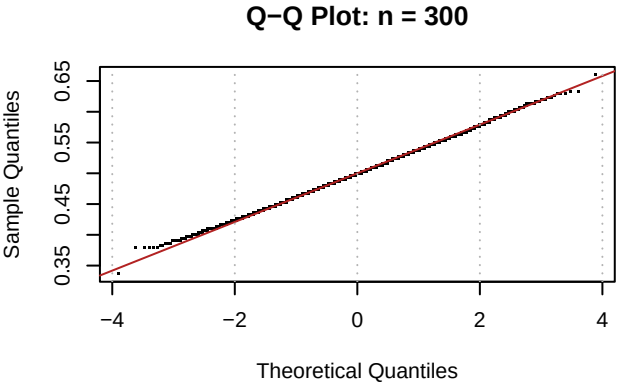

Sampling Distribution of Mean: n = 290
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

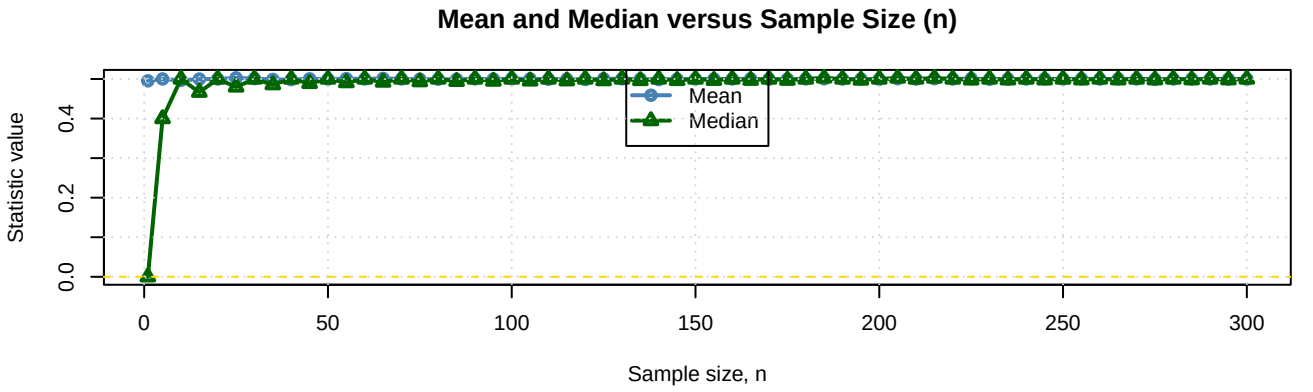

Q−Q Plot: n = 290
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

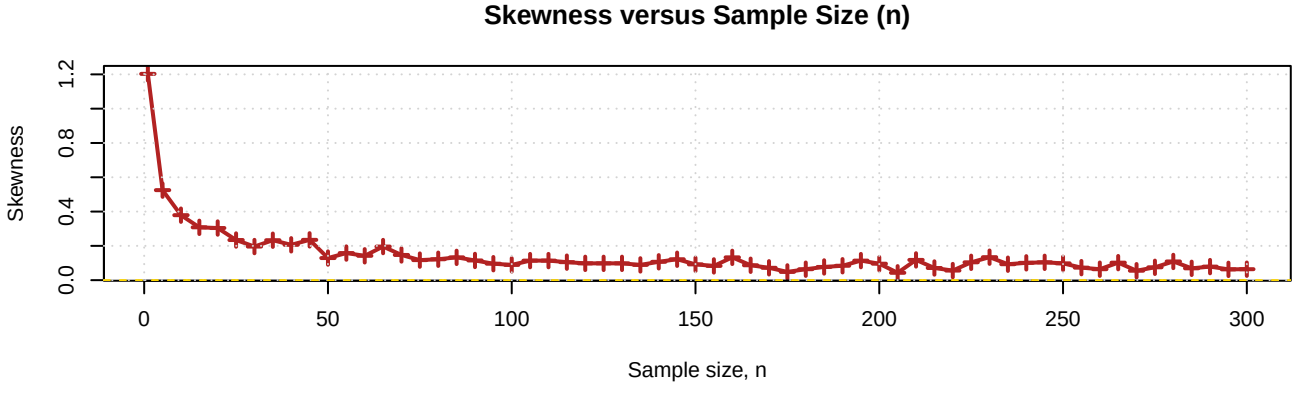

Sampling Distribution of Mean: n = 295
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

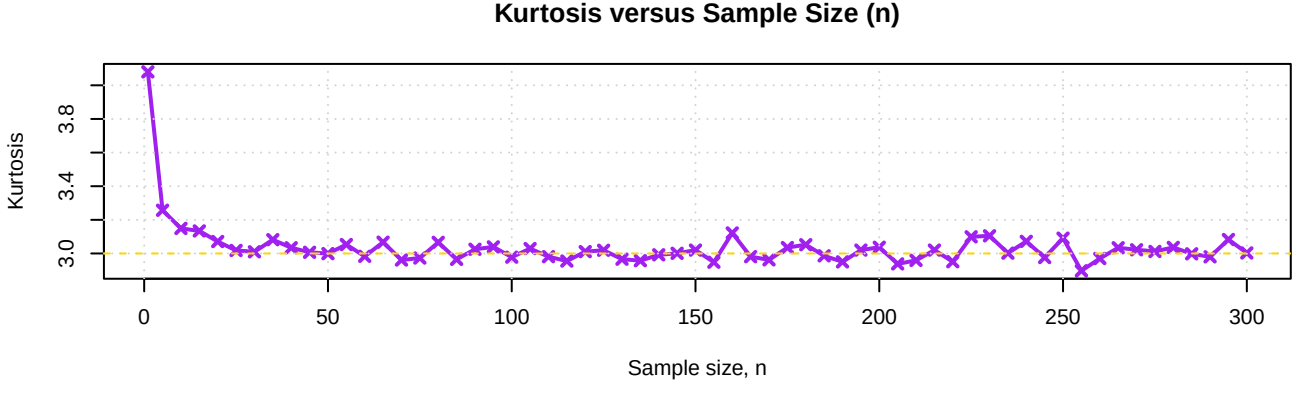

Q−Q Plot: n = 295
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 300
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean


Q−Q Plot: n = 300
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Statistic value
0.0
0.2
0.4
Mean
Median
0
50
100
150
200
250
300
Sample size, n


Skewness versus Sample Size (n)
Skewness
0.0
0.4
0.8
1.2
0
50
100
150
200
250
300
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
3.0
3.4
3.8
0
50
100
150
200
250
300
Sample size, n